

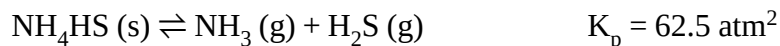
WORK SHEET - 01

CHEMICAL EQUILIBRIUM

[SINGLE CORRECT CHOICE TYPE]

- Q.1. At total equilibrium pressure P_1 atm and P_2 atm, N_2O_4 is dissociated to an extent of 33.33% and 50% respectively. Ratio of $\frac{P_1}{P_2}$ will be
- (A) $\frac{3}{8}$ (B) $\frac{4}{3}$ (C) $\frac{8}{3}$ (D) $\frac{3}{4}$
- Q.2. For the reaction : $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2$; $K = 10$ M
at 600 K 0.1 mol of $PCl_5(g)$ is placed in an evacuated 5L bulb. Then what would be the total pressure at equilibrium. (Take : $R = 0.08 \text{ L-atm-mol}^{-1} \text{ K}^{-1}$)
- (A) 9.6 atm (B) 0.96 atm (C) 4.8 atm (D) 1.92 atm
- Q.3. The dissociation equilibrium of a gas AB_2 can be represented as -
 $2AB_2(g) \rightleftharpoons 2AB(g) + B_2(g)$
The degree of dissociation is 'x' and is small compared to 1. The expression relating the degree of dissociation (x) with equilibrium constant K_p and total pressure P is -
- (A) $(2K_p/P)^{1/2}$ (B) K_p/P (C) $2K_p/P$ (D) $(2K_p/P)^{1/3}$
- Q.4. At a given temperature the following reaction is allowed to reach equilibrium in a vessel of volume V_1 litre. The degree of dissociation is α_1 . If at same temperature the volume of the reaction vessel is doubled (assuming the degrees of dissociation to be small) the new degree of dissociation shall be -
- $$PCl_5 \rightleftharpoons PCl_3 + Cl_2$$
- (A) $2\alpha_1$ (B) $\sqrt{\frac{\alpha_1}{2}}$ (C) $\sqrt{2\alpha_1}$ (D) $\sqrt{2} \alpha_1$
- Q.5. For the reversible system,
 $X(g) \rightleftharpoons Y(g) + Z(g)$
a certain quantity of X was heated at pressure P at a certain temperature. The equilibrium partial pressure of X was found to be $\frac{P}{7}$. What is the value of K_p at given temperature
- (A) $\frac{6P}{7}$ (B) $\frac{9P}{7}$ (C) $\frac{36P}{7}$ (D) 6P
- Q.6. The vapour density of N_2O_4 at a certain temperature is 30. What is the % dissociation of N_2O_4 at this temperature?
- (A) 53.3% (B) 106.6% (C) 26.7% (D) None
- Q.7. For the reaction : $A.3H_2O(s) \rightleftharpoons A. H_2O(s) + 2H_2O(g)$; $K_p = 9 \text{ atm}^2$
A 24.63 litre flask contains 1 mole of A. $H_2O(s)$ at 300 K. How many moles of $H_2O(g)$ should be added to the flask at the given temperature to drive the backward reaction for completion.
- (A) 3 moles (B) 5 moles (C) 8 moles (D) 2 moles

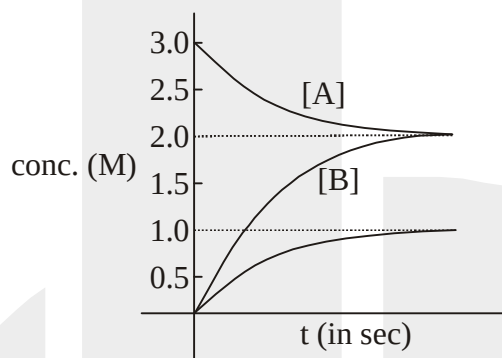
- Q.8 Ammonia at a pressure of 5 atm and H_2S gas at a pressure of 10 atm are introduced into an evacuated vessel.



The total pressure of gases after long time is :

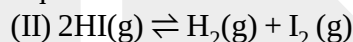
- (A) 15 atm (B) more than 15 atm (C) less than 15 atm (D) unpredicted

- Q.9 The progress of reaction : $\text{A(g)} \rightleftharpoons x\text{B(g)} + y\text{C(g)}$ with time is presented in figure. What is the value of K_c° at 300 K.



- (A) 1 (B) 2 (C) 3 (D) 4

- Q.10 Consider following reactions in equilibrium with equilibrium concentration 0.01 M of every species



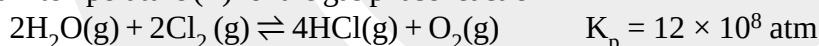
Extent of the reactions taking place is:

- (A) $\text{I} > \text{II} > \text{III}$ (B) $\text{I} < \text{II} < \text{III}$ (C) $\text{II} < \text{III} < \text{I}$ (D) $\text{III} < \text{I} < \text{II}$

- Q.11 A definite amount of solid NH_4HS is placed in a flask already containing ammonia gas at a certain temperature and 0.50 atm pressure. NH_4HS decomposes to give NH_3 and H_2S and at equilibrium total pressure in flask is 0.84 atm. The equilibrium constant for the reaction is :

- (A) 0.30 (B) 0.18 (C) 0.17 (D) 0.11

- Q.12 At certain temperature (T) for the gas phase reaction



If Cl_2 , HCl & O_2 are mixed in such a manner that the partial pressure of each is 2 atm and the mixture is brought into contact with excess of liquid water. What would be approximate partial pressure of Cl_2 when equilibrium is attained at temperature (T)?

[Given : Vapour pressure of water is 380 mm Hg at temperature (T)]

- (A) $3.6 \times 10^{-5} \text{ atm}$ (B) 10^{-4} atm (C) $3.6 \times 10^{-3} \text{ atm}$ (D) 0.01 atm

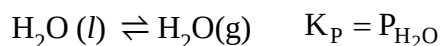
- Q.13 A vessel of 250 litre was filled with 0.01 mole of Sb_2S_3 and 0.01 mole of H_2 to attain the equilibrium at 440°C as $\text{Sb}_2\text{S}_3\text{(s)} + 3\text{H}_2\text{(g)} \rightleftharpoons 2\text{Sb(s)} + 3\text{H}_2\text{S(g)}$.

After equilibrium the H_2S formed was analysed by dissolving it in water and treating with excess of Pb^{2+} to give 1.195 g of PbS (Molecular weight = 239) precipitate.

What is value of K_c of the reaction at 440°C ?

- (A) 1 (B) 2 (C) 4 (D) None of these

Q.14 For the following Equilibria :



where $P_{\text{H}_2\text{O}} \longrightarrow$ Vapour pressure of $\text{H}_2\text{O}(g)$

By which of the following ways $P_{\text{H}_2\text{O}}$ can be changed

- (A) By adding more $\text{H}_2\text{O}(l)$ (B) By adding more $\text{H}_2\text{O}(g)$
(C) By changing temperature (D) All of the above

Q.15 For the reaction : $\text{A}(s) \rightleftharpoons \text{B}(g) + \text{C}(g)$.What will be the value of natural logarithm of ratio of total

pressure at 400K to that at 300K $\left[= \ln \frac{P_{400}}{P_{300}} \right]$ if $\Delta H = 16.628 \text{ kJ}$. (Given : $R = 8.314 \text{ J/K-mole}$)

- (A) 5/3 (B) 5/6 (C) 3/5 (D) 6/5

Q.16 At a equilibrium pressure of 3.3 atm N_2O_4 undergoes 10% decomposition to NO_2 . At same temperature what will be equilibrium pressure required for 20% dissociation.

- (A) 3.3 atm (B) 6.6 atm (C) 4 atm (D) 0.8 atm

[PARAGRAPH TYPE]

Paragraph for question nos. 17 to 19

Following reaction is at equilibrium in basic medium at 300 K.



Equilibrium concentration at 300 K are, $[\text{I}^-] = 0.1 \text{ M}$, $[\text{IO}_3^-] = 0.1 \text{ M}$

Given : $\Delta G_f^\circ (\text{I}^-, \text{aq}) = -50 \text{ kJ/mol}$

$$\Delta G_f^\circ (\text{IO}_3^-, \text{aq}) = -123.5 \text{ kJ/mol}$$

$$\Delta G_f^\circ (\text{H}_2\text{O}, l) = -233 \text{ kJ/mol}$$

$$\Delta G_f^\circ (\text{OH}^-, \text{aq}) = -150 \text{ kJ/mol}$$

$$R = \frac{25}{3} \text{ J mol}^{-1} \text{ K}^{-1} \log_{10} e = 2.3$$

Q.17 ΔG° of the reaction in KJ mol^{-1} is

- (A) -150.7 (B) -172.5 (C) 172.5 (D) 150.7

Q.18 Value of equilibrium constant is

- (A) 10^{25} (B) 10^{35} (C) 10^{30} (D) 10^{27}

Q.19 pH value at equilibrium is

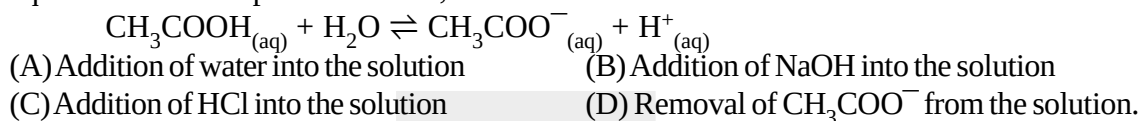
- (A) 8 (B) 6 (C) 9 (D) 5



Paragraph for question no. 20 to 22

According to Le Chatelier principle when an equilibrium is subjected to any external change, the equilibria/ reaction shifts to compensate the effect of the change. This principle helps in shifting the reaction towards appropriate directions so as to increase % yield of any reaction.

- Q.20 Which of the following changes can not cause an increase in extent of dissociation of CH_3COOH in its aqueous solution as per the reaction,



- Q.21 A,B,C & D are in equilibrium in a 2 litre container at 400 K & their moles are respectively 4, 5, 8 & 6. If the reaction involved is $3\text{A}_{(\text{g})} + 2\text{B}_{(\text{g})} \rightleftharpoons \text{C}_{(\text{g})} + 5\text{D}_{(\text{g})}$ then calculate equilibrium concentration of C when volume is increased to 10 litre.

(A) 4M (B) 0.8 M (C) 5 M (D) None of these

- Q.22 If it is given that conversion of graphite to diamond in an endothermic reaction & the conversion $\text{C}_{(\text{graphite})} \rightleftharpoons \text{C}_{(\text{diamond})}$ attains equilibria at 1.5×10^9 Pa at 300 K then comment at what pressure equilibria can be attained at 500 K.

(A) $P > 1.5 \times 10^9$ Pa (B) $P < 1.5 \times 10^9$ Pa
(C) $P = 1.5 \times 10^9$ Pa (D) At any pressure

Paragraph for question nos. 23 to 25

In a closed rigid vessel, N_2 and H_2 gases are taken in 9 : 13 mole ratio, by which the following equilibria are established :



At equilibrium, the total pressure is 14 atm, the partial pressure of ammonia is 2 atm and the partial pressure of hydrogen is 4 atm

- Q.23 The value of K_{P_1} is :

(A) $\frac{1}{80} \text{ atm}^2$ (B) $\frac{1}{80} \text{ atm}^{-2}$ (C) 80 atm^2 (D) $\frac{1}{40} \text{ atm}^{-2}$

- Q.24 The value of $\frac{1}{K_{P_2}}$ is :

(A) $\frac{3}{80} \text{ atm}^{-2}$ (B) $\frac{80}{3} \text{ atm}^2$ (C) $\frac{40}{3} \text{ atm}^2$ (D) $\frac{5}{3} \text{ atm}^{-2}$

- Q.25 The equilibrium partial pressure of N_2 is :

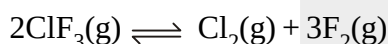
(A) 5 atm (B) 7 atm (C) 3 atm (D) 2 atm

[ASSERTION AND REASON]

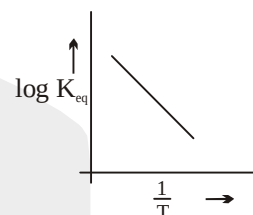
- Q.26 **Statement-1:** An increase in pressure (caused by decrease in volume) at equilibrium results in increase in molar concentration of each gaseous substance involved.
Statement-2: An increase in pressure (caused by decrease in volume) at equilibrium results in increase in number of moles of each gaseous substance involved.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

[MULTIPLE CORRECT CHOICE TYPE]

- Q.27 For the reaction



$\log K_{\text{eq}}$ v/s $\frac{1}{T}$ (where temperature is in K) curve is obtained as following



Which of the following change will increase the concentration of Cl_2 in an equilibrium mixture of Cl_2 , F_2 & ClF_3 :

- (A) Addition of inert gas at constant pressure
 (B) Increase in temperature at constant volume
 (C) Addition of catalyst at equilibrium
 (D) Removal of $\text{F}_2(\text{g})$ at equilibrium
- Q.28 For a gaseous reaction : $\text{A}_{(\text{g})} \rightleftharpoons 3\text{B}_{(\text{g})} + \text{C}_{(\text{g})}$, ΔH is positive and the reaction attains equilibrium at 1 bar total pressure and 400K. Identify the **incorrect** statements regarding the above reaction.
 (A) On increase of temperature, equilibrium will be shifted in forward direction.
 (B) When inert gas is introduced into a rigid container containing above equilibria equilibrium shifts towards left.
 (C) $\Delta G_{400}^\circ = 0$ for the above reaction.
 (D) If volume of vessel containing the above equilibria is increased without change in temperature then partial pressure of B decreases as compared to original equilibrium partial pressure of B.
- Q.29 Consider the equilibrium $\text{HgO}(\text{s}) + 4\text{I}^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{HgI}_4^{2-}(\text{aq}) + 2\text{OH}^-(\text{aq})$, which changes will decrease the equilibrium concentration of HgI_4^{2-}
 (A) Addition of 0.1 M HI (aq) (B) Addition of HgO (s)
 (C) Addition of $\text{H}_2\text{O}(\text{l})$ (D) Addition of KOH (aq)
- Q.30 For the reaction $\text{N}_2\text{O}_5(\text{g}) \rightarrow 2\text{NO}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g})$
 $\text{N}_2\text{O}_5(\text{g})$ decomposed at a constant volume and temperature. Initial pressure of N_2O_5 is 600 mm of Hg and total pressure at equilibrium is 960 mm of Hg. Assuming ideal behavior of gases, Select the correct option(s).
 (A) Involved reaction is intramolecular redox reaction
 (B) The mole fraction of NO_2 in the equilibrium mixture is 0.5
 (C) The equilibrium constant K_p for the above reaction is 540 mm of Hg
 (D) For the above reaction $K_c = K_p (\text{RT})^3$



Q.31 Select correct statement(s)

- (A) K_p depends upon equilibrium pressure
- (B) Dissociation of $\text{NH}_4\text{HS(s)}$ is suppressed at high pressure $[\text{NH}_4\text{HS(s)} \rightleftharpoons \text{NH}_3\text{(g)} + \text{H}_2\text{S(g)}]$
- (C) Low pressure is favourable for melting of ice
- (D) During dissociation of $\text{PCl}_5\text{(g)} \longrightarrow \text{PCl}_3\text{(g)} + \text{Cl}_2\text{(g)}$, M_{avg} decreases with progress of reaction

Q.32 Which of the following is/are correct statement(s)?

- (A) The vapour pressure of a solid is zero at absolute zero.
- (B) The maximum vapour pressure of a solid is its vapour pressure at the triple point.
- (C) The minimum vapour pressure of a liquid is its vapour pressure at the triple point.
- (D) The maximum vapour pressure of a liquid is its vapour pressure at critical temperature.

Q.33 The equilibrium between, gaseous isomers A, B and C can be represented as

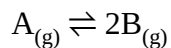
Reaction	Equilibrium constant
$\text{A (g)} \rightleftharpoons \text{B (g)}$	$K_1 = ?$
$\text{B (g)} \rightleftharpoons \text{C (g)}$	$K_2 = 0.4$
$\text{C (g)} \rightleftharpoons \text{A (g)}$	$K_3 = 0.6$

If one mole of A is taken in a closed vessel of volume 1 litre, then

- (A) $[\text{A}] + [\text{B}] + [\text{C}] = 1 \text{ M}$ at any time of the reactions
- (B) Concentration of C is 4.1 M at the attainment equilibrium in all the reactions
- (C) The value of K_1 is $\frac{1}{0.24}$
- (D) Isomer [A] is least stable as per thermodynamics.

[MATCHING THE COLUMN]

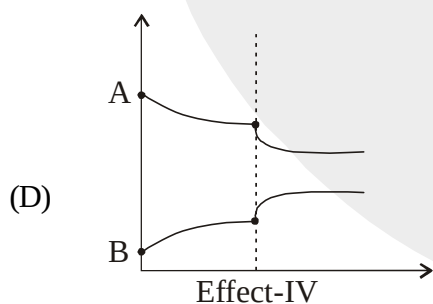
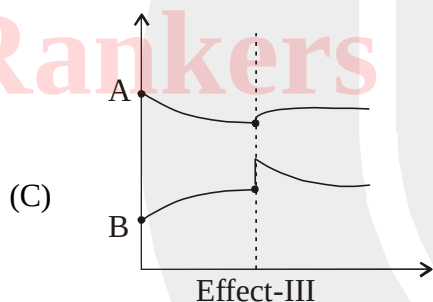
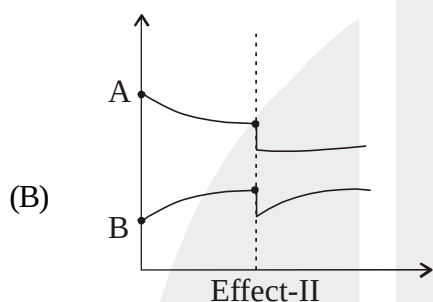
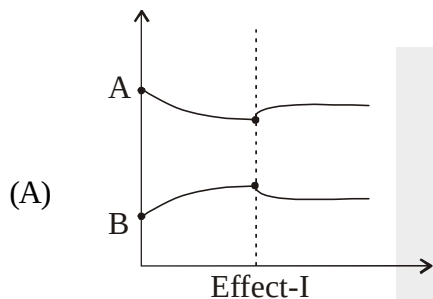
Q.34 For the endothermic reaction,



Certain changes were caused & the behaviour of equilibria was analysed by plotting changes in concentration v/s time. Match **List-I** with **List-II**.

List I

List II



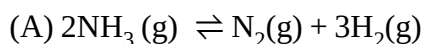
(P) Increase in temperature

(Q) Addition of inert gas at constant pressure.

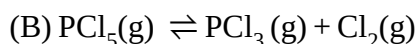
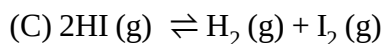
(R) Decrease in temperature

(S) Addition of $B_{(g)}$ at equilibrium

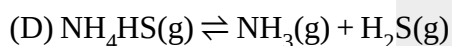
Q.35

Column I**Column II**

(P) Introduction of hydrogen gas at constant volume shift equilibrium back.

(Q) α increases on increasing temperature.

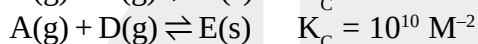
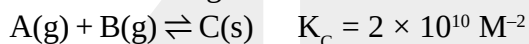
(R) On adding inert gas at constant volume, equilibrium state does not change



(S) On increasing the pressure, concentration of all reactants increases

(T) α is independent of equilibrium pressure.**[NUMERICAL INTEGER TYPE QUESTIONS]**

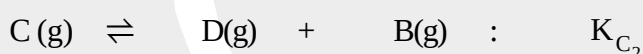
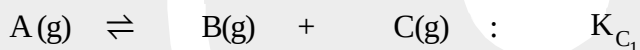
Q.36 5 moles each of 'A', 'B' and 'D' are added to a 1 litre container. Calculate the number of moles of 'B' at equilibrium if following reaction occurs.



[Fill your answer by multiplying it with 150]

Q.37 PCl_5 dissociates according to the reaction $\text{PCl}_5 \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$. At 523 K, $K_p = 1.78 \text{ atm}$. Find the density of the equilibrium mixture at a total pressure of 1 atm.

Q.38 When 1 mole of A(g) is introduced in a closed rigid 1 litre vessel maintained at constant temperature the following equilibria are established.



The pressure at equilibrium is twice the initial pressure. Calculate the value of $\frac{K_{C_2}}{K_{C_1}}$ if $\frac{[\text{C}]_{\text{eq}}}{[\text{B}]_{\text{eq}}} = \frac{1}{5}$

ANSWER KEY

Q.1 (C)	Q.2 (D)	Q.3 (D)	Q.4 (D)	Q.5 (C)
Q.6 (A)	Q.7 (B)	Q.8 (A)	Q.9 (B)	Q.10 (B)
Q.11 (D)	Q.12 (C)	Q.13 (A)	Q.14 (C)	Q.15 (B)
Q.16 (D)	Q.17 (B)	Q.18 (C)	Q.19 (A)	Q.20 (C)
Q.21 (D)	Q.22 (B)	Q.23 (B)	Q.24 (B)	Q.25 (A)
Q.26 (C)	Q.27 (BD)	Q.28 (BC)	Q.29 (CD)	Q.30 (AB)
Q.31 (BD)	Q.32 (ABCD)	Q.33 (ACD)		
Q.34 (A) R (B) Q (C) S (D) P				
Q.35 (A) P, Q, R, S ; (B) Q, R, S ; (C) P, Q, R, S, T ; (D) Q, R, S				
Q.36 (0250)	Q.37 (2.7)	Q.38 (4)		

